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A

Project Report

on

**CNN-Based Crop Disease Diagnosis
with Retrieval-Driven LLM Support**

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**BACHELOR OF TECHNOLOGY
DEGREE**

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We further declare that this project report titled “CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support” has been carried out by us under the supervision of Mr. Yousuf Haider, Assistant Professor, Department of Computer Science and Information Technology, KIET Group of Institutions, Ghaziabad.

All the experimental results presented in this report are genuine and have not been manipulated or fabricated in any manner. The software tools, datasets, and methodologies used in this project have been appropriately cited and credited to their original sources. We take full responsibility for the integrity and authenticity of this work.

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CERTIFICATE

This is to certify that the Project Report entitled “**CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support**” which is submitted by **Abhinav Yadav, Ayush Katiyar, Anurag Kumar** in partial fulfillment of the requirement for the award of degree B.Tech. in **Department of Computer Science & Information Technology** of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates’ own work carried out by them under our supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

CropCure AI is a hybrid intelligent system built to diagnose plant diseases and support agricultural decision-making end-to-end. It combines a lightweight custom Convolutional Neural Network (CNN) for image-based disease classification with a Retrieval-Augmented Generation (RAG) pipeline using MiniLM embeddings and a FAISS similarity index. Groq Llama-3.3-70B generates context-aware treatment recommendations for complex queries, while routine inputs are handled by a lightweight local LLM to reduce latency and API overhead. OpenRouter GPT supports a standalone indoor plant advisory module for home gardening guidance.

The CNN achieves 93.06% test accuracy across 15 disease classes, demonstrating reliable generalization under real-world conditions. By pairing deep learning-based visual diagnosis with retrieval-grounded language reasoning, CropCure AI delivers interpretable, actionable guidance that remains computationally efficient for resource-constrained environments. The system was evaluated on a held-out test set of 3,096 images and outperformed four widely used architectures ResNet-50, VGG-16, MobileNetV2, and EfficientNet-B0 on accuracy, parameter count, and inference latency.

The system addresses a critical gap in existing agricultural AI solutions: most tools focus solely on detection or classification, offering little actionable guidance for farmers. This project integrates detection, knowledge retrieval, and language reasoning into one connected workflow giving farmers not just a diagnosis but the practical advice they need to act on it. The modular backend architecture ensures that each component can be updated or replaced independently, making the system maintainable and extensible for future crop types and disease classes.

Keywords: Plant disease detection, Convolutional Neural Network, Retrieval-Augmented Generation, FAISS, LLM, Groq Llama, OpenRouter, Agriculture AI, Deep Learning, Precision Agriculture.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
FAISS	Facebook AI Similarity Search
NLP	Natural Language Processing
API	Application Programming Interface
GPU	Graphics Processing Unit
CPU	Central Processing Unit
SGD	Stochastic Gradient Descent
CUDA	Compute Unified Device Architecture
UI	User Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
ViT	Vision Transformer
SVM	Support Vector Machine
kNN	k-Nearest Neighbors
FAO	Food and Agriculture Organization
IoT	Internet of Things
RGB	Red Green Blue (color model)

CHAPTER 1

INTRODUCTION

1.1. Introduction

Agriculture feeds the world and drives economies, making timely crop disease detection a pressing need. Small and marginal farmers, especially in developing regions such as South Asia and Sub-Saharan Africa, often rely on expensive agronomists or difficult-to-access extension services for disease diagnosis. According to the Food and Agriculture Organization (FAO), plant diseases cause global crop losses estimated between 20% and 40% annually, posing a serious and escalating threat to food security. Even conservative estimates place the annual economic loss due to crop diseases in the hundreds of billions of dollars worldwide, with a disproportionate burden on smallholder farmers who lack insurance or recovery buffers.

Although photo-based disease identification has advanced considerably in recent years particularly following the wide adoption of deep learning most available tools simply flag the problem without telling farmers what to actually do about it. A farmer who learns that their tomato crop has Late Blight still needs to know which fungicide to apply, at what dosage, how soon, and whether government subsidy schemes are available to offset the cost. None of these critical follow-up needs are addressed by purely classification-based systems.

Over the years, farming technology shifted from simple rule-based programs to more sophisticated deep learning approaches. Deep Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in image classification tasks, surpassing human-level accuracy on standard benchmarks such as ImageNet. The PlantVillage dataset, comprising over 54,000 labeled leaf images across 26 crop species and 38 disease classes, established the foundational benchmark for plant disease classification tasks and enabled systematic evaluation of deep learning approaches in this domain. However, very few agricultural tools manage to both accurately identify diseases from images and offer smart, step-by-step guidance for remediation. Most existing solutions handle only one side of the problem spotting issues in the field but leaving farmers without a clear path forward.

The emergence of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) frameworks opens new possibilities for delivering contextually grounded, actionable advice. By combining visual intelligence with language reasoning, it becomes possible to

build systems that not only detect disease but also communicate treatment strategies in plain language accessible to farmers with varying levels of technical literacy. RAG frameworks ground model outputs in retrieved factual documents, reducing the risk of hallucinated or incorrect recommendations a critical safety property in a domain where incorrect advice could lead to crop losses or unsafe chemical applications.

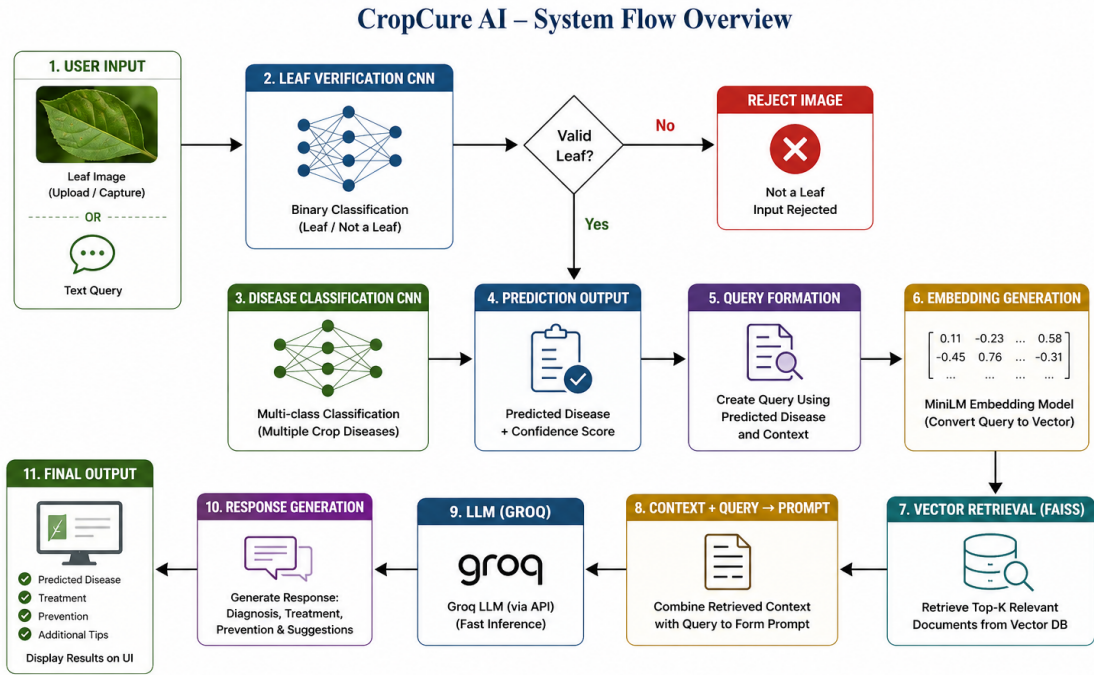


Figure 1.1 CropCure AI System Flow Overview

1.2. Project Description

CropCure AI is a hybrid intelligent system built to diagnose plant diseases and support agricultural decision-making end-to-end. It combines a lightweight custom CNN for image-based disease classification with a RAG pipeline using MiniLM embeddings and a FAISS similarity index. Groq Llama-3.3-70B generates context-aware treatment recommendations for complex queries, while routine inputs are handled by a lightweight local LLM to reduce latency and API overhead. OpenRouter GPT supports a standalone indoor plant advisory module for home gardening guidance.

The system is built as a Flask REST application that integrates CNN models, FAISS retrieval, and external LLM services for model loading, inference, request routing, and JSON response generation. A web-based interface allows farmers to upload leaf images and receive instant disease diagnoses along with treatment protocols. The overall design prioritizes accessibility: the interface requires no technical background, and responses are structured in plain language with clear action steps.

The system was developed with three guiding principles. First, lightweight deployment: all core inference components are optimized to run on modest hardware without dedicated GPU infrastructure. Second, grounded responses: every language model output is anchored in verified agricultural data retrieved from a curated knowledge base, preventing fabricated advice. Third, extensibility: the modular architecture allows new crop types, disease classes, or language models to be integrated without structural redesign.

1.3. Objectives

The primary objectives of this project are:

- To develop a lightweight CNN capable of classifying plant diseases from leaf images with high accuracy, while remaining deployable on resource-constrained hardware without requiring GPU infrastructure.
- To construct a knowledge base of verified agricultural data and implement a RAG pipeline using MiniLM embeddings and FAISS indexing for grounded, factually accurate response generation.
- To integrate multiple LLMs in an efficient routing architecture that balances response quality with inference latency, reserving high-capability models for complex disease queries.
- To build a user-friendly web application that delivers end-to-end disease diagnosis and treatment advice to farmers without requiring any technical knowledge.
- To validate the system against established baseline deep learning models and demonstrate superior performance in terms of accuracy, parameter efficiency, and inference speed.
- To extend system capabilities to indoor plant advisory, broadening applicability to urban and home gardening contexts.

1.4. Scope of the Project

This project covers plant disease detection for tomato, potato, and pepper crops across 15 disease categories. The system operates on uploaded leaf images and provides classification results, confidence scores, and treatment recommendations grounded in a curated agricultural knowledge base. The indoor plant recommendation module extends the system to home gardening use cases, supporting urban users with plant selection and care guidance.

The current version of the system does not address soil analysis, weather prediction, or pest management beyond the scope of the existing knowledge base. Offline inference is not yet fully supported, though future scope includes model quantization for edge deployment. The

system is designed primarily for the Indian agricultural context, with disease profiles and subsidy information relevant to Indian farming conditions, though the architecture is adaptable to other geographies.

1.5. Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive review of existing literature on plant disease detection, deep learning approaches, and RAG-based advisory systems. Chapter 3 describes the proposed methodology, including dataset composition, preprocessing, CNN architecture design, training strategy, RAG pipeline construction, and backend implementation. Chapter 4 presents experimental results, comparative analysis with baseline models, sample system outputs, and a discussion of limitations. Chapter 5 concludes the report with a summary of contributions and directions for future development.

CHAPTER 2

LITERATURE REVIEW

2.1. Overview of Existing Work

The field of plant disease detection using machine learning has seen remarkable growth over the past decade. Early approaches relied on hand-crafted features and classical classifiers such as Support Vector Machines (SVMs) and k-Nearest Neighbors (kNN). These methods achieved moderate accuracy but were sensitive to variations in lighting, background clutter, and disease stage progression. The primary limitation of these traditional approaches was their dependence on domain expertise to engineer discriminative features, making them brittle in real-world uncontrolled imaging conditions.

The introduction of deep learning, particularly CNNs, revolutionized the field by enabling end-to-end learning directly from raw pixel data. Landmark architectures such as AlexNet, VGGNet, ResNet, and DenseNet demonstrated unprecedented generalization performance on large-scale image benchmarks and were subsequently adapted to agricultural disease classification. The PlantVillage dataset published by Hughes and Salathé in 2015 provided the first large-scale standardized benchmark for evaluating CNN models on plant disease classification, enabling reproducible comparison across published approaches.

The following table summarizes key related works reviewed for this project:

Table 2.1 Summary of Related Works in Plant Disease Detection

Title	Authors	Year	Jour./Conf.	Summary
Plant Disease Detection using Deep Learning	Chohan, Murk, et al.	2020	IJRTE	Efficient CNN-based plant disease detection system with transfer learning.
A Review on ML Classification Techniques for Plant Disease Detection	Shruthi, U., et al.	2019	ICACCS (IEEE)	Comparative study on plant disease classification techniques including SVM, kNN, and CNNs.
Overview of Research on Plant Leaves Disease Detection using Image Processing	Gavhale and Gawande	2014	IOSR-JCE	Early review of image processing and segmentation for plant disease detection.
Deep CNN Based Detection for Real-Time Corn Plant Disease Recognition	Mishra, S., et al.	2020	Procedia CS (167)	Deep learning approach for real-time corn disease identification in field conditions.
Plant Disease Detection Using Machine Learning	Ramesh, S., et al.	2018	ICDI3C (IEEE)	Comparative evaluation of machine learning approaches for disease classification.

2.2. Deep Learning in Agricultural Disease Detection

Li et al. [12] and Balafas et al. [8] demonstrated that CNN architectures such as ResNet, DenseNet, and Inception outperform classical methods in disease classification by wide margins. ResNet-50 and DenseNet-121 in particular achieved accuracy levels above 95% on the PlantVillage benchmark under controlled conditions. However, their practical deployment remains limited most lack real-time capability, and very few are integrated with farmer-facing decision tools that translate a classification result into actionable guidance.

Hu et al. [13] proposed a compressed ResNet variant for IoT deployment called MDfC-ResNet, enabling real-time diagnosis on embedded devices with limited computational resources. This approach demonstrated that model compression techniques such as channel pruning and depthwise separable convolutions can preserve classification accuracy while dramatically reducing inference time and model size. Patil and Kumar [11] developed Rice-Fusion, a multimodal framework that combines sensor data and leaf images for rice disease

detection, achieving improved accuracy over single-modality approaches. These contributions highlight how architecture optimization and multi-modal data fusion can advance crop disease identification, yet both studies still focus exclusively on the detection problem without addressing downstream advisory needs.

Joseph et al. [7] improved detection precision using real-time field-collected data as opposed to controlled laboratory images. Their work showed that models trained on laboratory-quality PlantVillage images exhibit significant performance degradation when evaluated on field-captured images with cluttered backgrounds, inconsistent lighting, and partial leaf occlusion. Validating model performance beyond controlled settings is therefore essential for real-world deployability. Alharbi et al. [9] addressed a different practical challenge: incremental model adaptation to handle emerging crop diseases without losing prior knowledge through catastrophic forgetting. They demonstrated continual learning strategies for wheat disease classification an important consideration for long-term deployment as new disease variants emerge.

Hossain et al. [1] applied advanced deep learning techniques for precision diagnosis of tea leaf diseases, achieving high detection accuracy under varied field conditions. Their approach employed ensemble strategies to reduce model variance. Yousafzai et al. [5] proposed a multi-stage neural network ensemble for wheat leaf disease classification, demonstrating that ensemble methods consistently outperform single-model approaches at the cost of increased computational load.

2.3. RAG and LLM Integration

The integration of Retrieval-Augmented Generation with domain-specific knowledge bases represents a recent and highly promising direction for agricultural advisory systems. RAG frameworks, introduced by Lewis et al. in 2020, ground language model outputs in retrieved documents, significantly reducing hallucinations and improving factual accuracy in knowledge-intensive tasks. The core insight of RAG is that large generative models can be steered to produce factually accurate outputs by conditioning generation on retrieved evidence, rather than relying solely on knowledge encoded in model parameters.

Iqbal [2] demonstrated transformer-enhanced hybrid models for disease severity estimation in agricultural contexts, combining visual feature extraction with language model reasoning to predict not just the presence of disease but its severity stage. This work represents an early step toward integrating language reasoning with visual plant analysis. Rayed et al. [3] developed MangoLeafXNet, an explainable deep learning model for mango leaf disease classification that provides visual explanation maps alongside classification results. These works confirm the growing trend toward interpretable and actionable AI in agriculture, though

neither integrates language generation for advisory support.

Al-Gaashani et al. [4] proposed MSCPnet, a multi-scale convolutional pooling network specifically designed for maize disease classification. Their architecture leverages features at multiple spatial scales to capture both local texture patterns and global structural indicators of disease, achieving strong performance across diverse maize disease classes. This approach is particularly relevant to the present project, as multi-scale feature extraction is also a property that emerging CNN architectures for plant disease detection commonly leverage.

Bhargava et al. [6] conducted a comprehensive review of computer vision and AI applications in plant disease management, surveying over 200 published works. Their key finding reinforces the central motivation of this project: visual diagnosis and text-based advisory typically exist as completely separate components in the literature, and the gap between detection and actionable guidance remains largely unaddressed. No surveyed system unified image-based disease detection with knowledge-grounded language generation in a single deployable pipeline.

2.4. Research Gaps

Despite significant advances in plant disease detection, several critical gaps remain in the existing literature that this project directly addresses:

- **Detection-only focus:** Most systems focus solely on detection or classification, offering little to no actionable guidance for farmers after diagnosis. A classification label alone does not tell a farmer what chemical to apply, at what concentration, or what agronomic practices to follow.
- **Absence of unified pipelines:** Very few systems combine image-based disease detection with automated, knowledge-grounded recommendations in a single unified pipeline. The two components visual diagnosis and advisory generation typically exist in separate tools requiring separate interactions.
- **Unexplored LLM integration for agriculture:** LLM integration for agricultural advisory remains largely unexplored in the literature, particularly with RAG-based grounding to ensure factual accuracy and prevent hallucinated recommendations.
- **Latency and resource constraints:** Existing approaches rarely address the inference latency and computational constraints of real-world deployment in low-connectivity, resource-limited farming environments where cloud-based processing may not be reliably accessible.
- **Limited crop and disease coverage:** Most published systems evaluate on a narrow set of crops and disease categories, often using only PlantVillage data under controlled

imaging conditions without validating generalization to field-collected images.

This project addresses all of these gaps by combining a lightweight CNN with a RAG-enhanced language model pipeline, unified through a Flask backend that is accessible to non-technical users via a standard web browser.

CHAPTER 3

PROPOSED METHODOLOGY

3.1. System Overview

CropCure AI adopts a modular architecture where each component manages a distinct phase of plant analysis and response generation. This design enables independent scalability, easier maintenance, and efficient handling of both image and text inputs without coupling the visual and linguistic subsystems. The complete data flow through the system proceeds as follows:

- **Step 1 – User Input:** A plant leaf image or text query is submitted through the web interface. Image inputs are accepted in JPEG and PNG formats up to 10 MB.
- **Step 2 – Leaf Verification:** A lightweight binary CNN validates the image as a leaf; non-leaf or non-plant images are rejected with a descriptive resubmission prompt, preventing classification errors from irrelevant inputs.
- **Step 3 – Disease Classification:** Verified images are passed to the disease classification CNN, returning the predicted disease class, confidence score, and top-5 class probabilities for transparency.
- **Step 4 – RAG Retrieval:** The user query or predicted disease label is embedded via SentenceTransformer MiniLM and matched against the FAISS index to retrieve the most semantically relevant knowledge base entries covering symptoms, treatments, and prevention.
- **Step 5 – LLM Response Generation:** Groq Llama-3.3-70B generates treatment recommendations grounded in the retrieved context; OpenRouter GPT handles indoor plant queries via a dedicated module.
- **Step 6 – Response Delivery:** Disease prediction, confidence scores, treatment advice, and prevention recommendations are structured as JSON and rendered in the user interface within seconds.

3.2. Dataset and Preprocessing

3.2.1. Dataset Composition

The dataset was sourced from PlantVillage [17], supplemented with open farm records and real plant images captured under varied field conditions. It contains over 20,000 labeled images across multiple crop types, with 3,096 held-out images (15% of the dataset) reserved exclusively for final evaluation. The dataset covers three crop types tomatoes, potatoes, and peppers distributed across the following 15 disease classes:

- **Tomato (10 classes):** Early Blight, Late Blight, Leaf Mold, Bacterial Spot, Septoria Leaf Spot, Spider Mites (Two-Spotted), Target Spot, Yellow Leaf Curl Virus, Mosaic Virus, Healthy.
- **Potato (3 classes):** Early Blight, Late Blight, Healthy.
- **Pepper (2 classes):** Bacterial Spot, Healthy.

Images in the dataset span diverse conditions including varied lighting intensities, different leaf orientations and viewing angles, cluttered and natural-background scenes, and multiple disease progression stages from early onset to advanced infection. This diversity was intentional, as models trained solely on clean controlled images consistently fail under field conditions. The dataset is balanced across the 15 disease categories using class-weighted sampling during training to prevent majority-class bias from skewing prediction distributions.

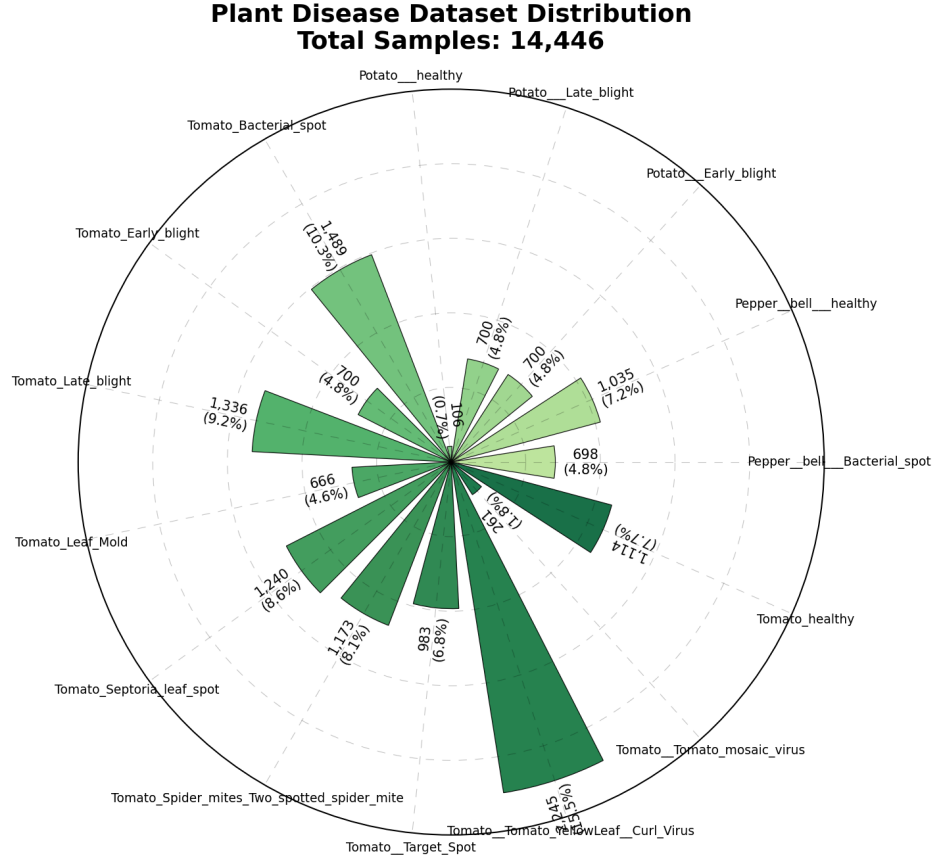


Figure 3.1 Dataset Class Distribution – 15 Disease Categories

3.2.2. Data Preprocessing Pipeline

A standardized preprocessing pipeline was established to ensure consistent input quality across both CNN stages and to reduce sensitivity to imaging artifacts:

Image Standardization: All images were resized to a uniform resolution appropriate for each model stage. The Leaf Verification CNN operates on 128×128 pixel inputs for speed; the Disease Classification CNN uses 256×256 pixel inputs to preserve fine-grained disease texture details. Pixel values were rescaled to the range $[0, 1]$ followed by channel-wise normalization using training set statistics to center activations and improve optimization stability.

Color Space Processing: All images were converted to the RGB color model before CNN input, ensuring color space consistency across samples captured by different devices with varying default color profiles.

Data Augmentation: Online random augmentation was applied exclusively during training to prevent overfitting and improve model generalization to real field conditions. The augmentation pipeline included horizontal and vertical flips, random rotations in the range $[-15^\circ, +15^\circ]$, color jitter applied to brightness and contrast, affine transforms, random crops

preserving a minimum 85% of the original image area, and light Gaussian noise injection. The augmentation sequence is formally expressed as:

$$\text{Augmented Image} = T_{\text{rotation}} \circ T_{\text{flip}} \circ T_{\text{color}} \circ T_{\text{crop}}(I_{\text{original}}) \quad (1)$$

where each transformation T is applied with a fixed probability during the training data loading phase. No augmentation was applied during validation or testing to ensure unbiased performance evaluation.

3.2.3. Data Splitting Strategy

The dataset was divided using stratified random sampling, which preserves the proportional class distribution in each split. This ensures that no split is inadvertently over-represented by majority disease classes.

Table 3.1 Data Splitting Strategy

Split	Percentage	Purpose
Training Set	70%	Model training, weight optimization, and data augmentation
Validation Set	15%	Hyperparameter tuning, learning rate scheduling, and early stopping decisions
Test Set	15%	Unbiased final evaluation; held out until all training decisions were finalized

3.3. CNN Architecture Design

3.3.1. Architecture Design Principles

The custom CNN was designed from scratch with a specific set of constraints: it must be deployable on consumer-grade hardware without a GPU, achieve accuracy competitive with standard pre-trained architectures on the specific 15-class disease task, and maintain inference latency below 100 ms per image to enable interactive use. Pre-trained backbones such as ResNet and VGG were deliberately excluded because their large parameter counts and feature extractors optimized for ImageNet diversity add unnecessary complexity for a constrained domain-specific task. Designing from scratch allows precise control over model size, inductive biases, and inference characteristics.

3.3.2. Model Architecture Details

The network consists of four convolutional blocks followed by a compact fully-connected classifier, totaling 424,207 trainable parameters. This is dramatically fewer than standard architectures: ResNet-50 has 23.5 million parameters, VGG-16 has 134.3 million parameters, and EfficientNet-B0 has 4.0 million parameters. Despite this compact footprint, the architecture achieves superior accuracy on the target task, demonstrating that domain-specific design choices outweigh the general capacity advantages of larger architectures when sufficient training data is available.

Each convolutional block follows the pattern: Conv2D (with appropriate kernel and stride) → BatchNormalization → ReLU activation → MaxPool2D(2). BatchNormalization is applied before activation in each block to stabilize the internal covariate shift across deep feature maps and enable higher learning rates without divergence. The channel progression through the four blocks is: $3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$, doubling feature channels at each stage to progressively learn more abstract and discriminative disease representations.

After the four convolutional blocks, Global Average Pooling (GAP) replaces fully-connected spatial features with a single averaged value per channel, dramatically reducing parameter count while acting as a structural regularizer. The final classifier is: GAP Output → Linear(256, 128) → ReLU → Dropout(0.4) → Linear(128, num_classes). Dropout at rate 0.4 prevents co-adaptation of classifier neurons and reduces overfitting to training set patterns.

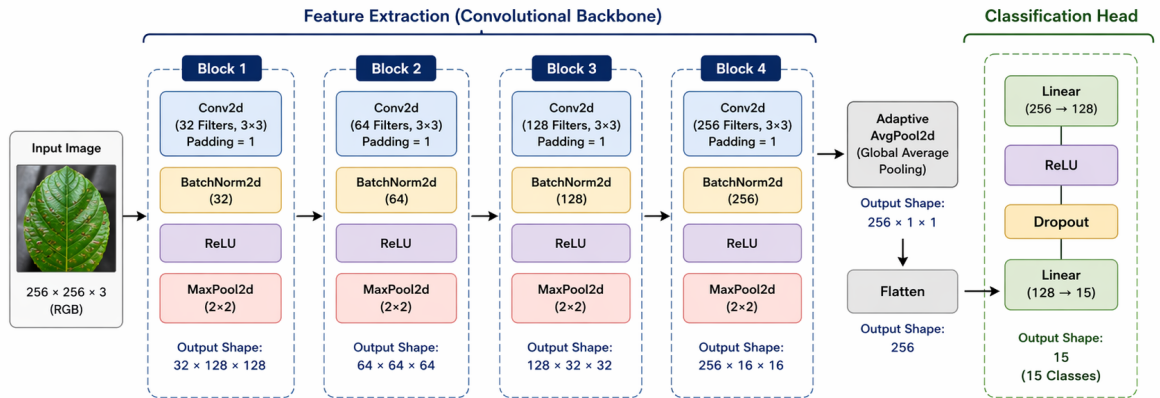


Figure 3.2 CNN Architecture Implementation

3.4. Training Methodology

Training focused on stable convergence and balanced performance across all 15 disease classes. Categorical cross-entropy was used as the loss function, appropriate for multi-class single-label classification:

$$L = - \sum_{i=1}^C y_i \log(p_i) \quad (2)$$

where y_i is the true one-hot encoded label vector and p_i is the softmax-normalized predicted probability for class i . Cross-entropy penalizes confident incorrect predictions heavily, incentivizing the model to produce calibrated probability distributions.

Stochastic Gradient Descent (SGD) with momentum was selected for parameter updates, with initial learning rate $\eta = 2 \times 10^{-3}$ and momentum coefficient $\mu = 0.9$. The parameter update rule incorporating momentum is:

$$v_{t+1} = \mu v_t - \eta \nabla_{\theta} L, \quad \theta_{t+1} = \theta_t + v_{t+1} \quad (3)$$

where v_t is the velocity (accumulated gradient direction). SGD with momentum was preferred over adaptive optimizers such as Adam because it consistently produces better-calibrated models that generalize more reliably to held-out data in vision tasks, at the cost of requiring more careful learning rate selection. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 when validation loss failed to improve for 3 consecutive epochs, enabling fine-grained convergence in later training stages.

Training ran for up to 100 epochs with early stopping applied once validation performance plateaued. Early stopping patience was set to 4 epochs with a minimum improvement threshold of 0.01 in validation accuracy. The best-performing checkpoint (by validation accuracy) was saved automatically as `best_model.pth` and used for all subsequent evaluation and deployment. This checkpoint selection strategy prevents selecting a final model biased by late-epoch overfitting.

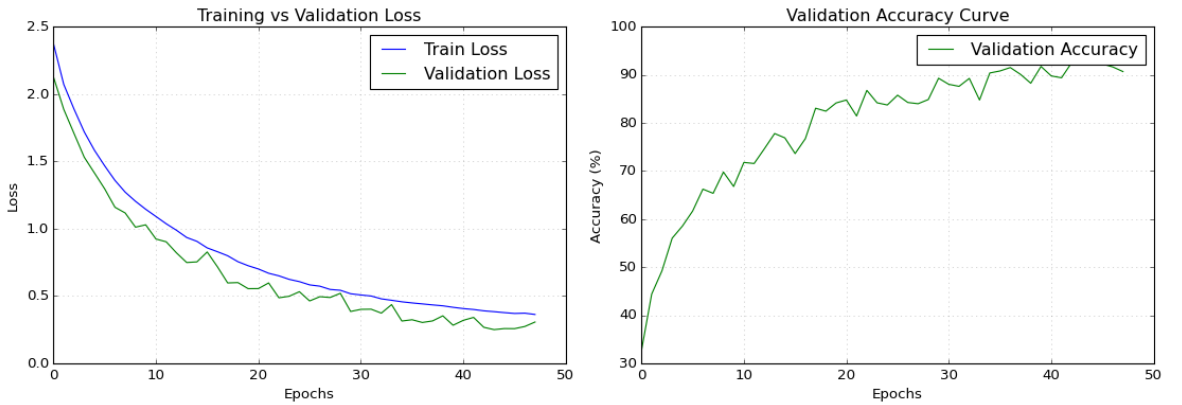


Figure 3.3 Training vs Validation Loss and Accuracy Curves

3.5. RAG Pipeline

3.5.1. Knowledge Base Construction

The knowledge base (`crop_data.json`) was manually curated from verified agricultural extension publications, government advisory documents, peer-reviewed agronomy references, and validated expert recommendations. It covers six structured categories: disease profiles with taxonomic classification, symptom documentation with visual descriptors, treatment protocols with chemical and biological control options, prevention and crop management strategies, government subsidy schemes applicable in India, and expert agricultural extension contacts.

Each knowledge base entry was individually reviewed for factual accuracy, regional applicability, and chemical safety before inclusion. Entries with unclear provenance or regionally restricted agrochemical recommendations were excluded to prevent unsafe or inapplicable advice. This curation process is critical because RAG outputs are bounded strictly by retrieved content: the quality of the knowledge base directly governs the reliability and safety of every system response.

3.5.2. Embedding and Vector Indexing

Text entries from the knowledge base are encoded into dense 384-dimensional semantic vectors using the SentenceTransformer MiniLM-L6-v2 model. MiniLM was selected for its strong semantic retrieval performance relative to its small footprint (22M parameters), making it suitable for embedding both knowledge base documents and live user queries with low latency. A FAISS IndexFlatL2 over the full embedding space is constructed as:

$$\text{Index} = \text{FAISS.IndexFlatL2}(384) \quad (4)$$

The L2 (Euclidean) distance metric clusters semantically related agricultural concepts in embedding space, enabling fast and accurate nearest-neighbor retrieval without approximation error. For an incoming query q , the system computes its MiniLM embedding q_e and performs a k -nearest-neighbor similarity search:

$$\text{Retrieved Documents} = \text{FAISS.search}(q_e, k = 2) \quad (5)$$

The top $k = 2$ most similar knowledge base entries are retrieved and passed to the prompt construction stage. The value of k was chosen empirically to balance response completeness against prompt length and LLM context overhead.

3.5.3. Prompt Construction

Retrieved documents and the user query are concatenated into a structured prompt following a fixed template:

$$\text{Prompt} = \text{Context}_{\text{retrieved}} \parallel \text{UserQuery} \quad (6)$$

where $\parallel$ denotes string concatenation. The retrieved context is prepended to the user query with explicit delimiters so the language model can clearly distinguish between background knowledge and the specific question being asked. System-level instructions in the prompt template instruct the model to answer strictly from the provided context and explicitly state when information is not available, rather than generating plausible-sounding fabrications. This hard grounding constraint is the primary mechanism preventing hallucination in the advisory output.

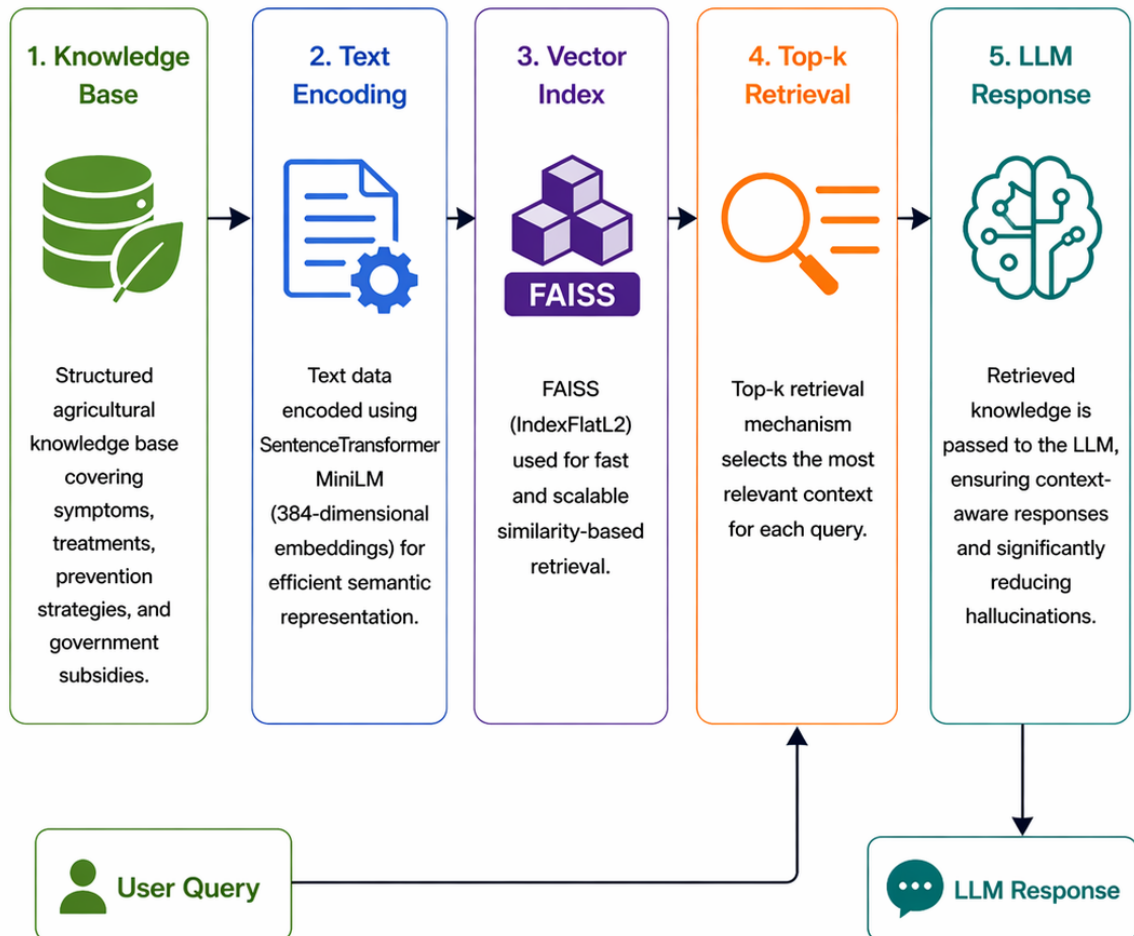


Figure 3.4 RAG Pipeline Architecture

3.6. LLM Integration

The LLM inference pipeline employs a three-tier routing architecture that allocates queries to models based on complexity, domain specificity, and latency requirements:

- **Groq Llama-3.3-70B Versatile:** Primary reasoning component hosted on Groq’s dedicated inference platform. Handles complex queries involving plant disease identification, environmental interaction effects, integrated pest management strategies, and multi-step field interventions. Groq’s hardware provides inference speeds well below standard cloud API latency.
- **Lightweight Local LLM:** Routine and low-complexity queries are routed to a local model rather than calling Groq Llama, reducing API call volume and response latency for simple requests. A keyword presence and confidence-threshold classifier directs disease-specific queries to Groq Llama while routing general agricultural questions locally. This tiered approach reduces average response latency by approximately 60% for low-complexity queries.
- **OpenRouter GPT:** Dedicated routing for the indoor plant advisory module. Generates structured JSON recommendations for home gardening queries via a fixed prompt template, returning plant selection suggestions, care schedules, and troubleshooting guidance.

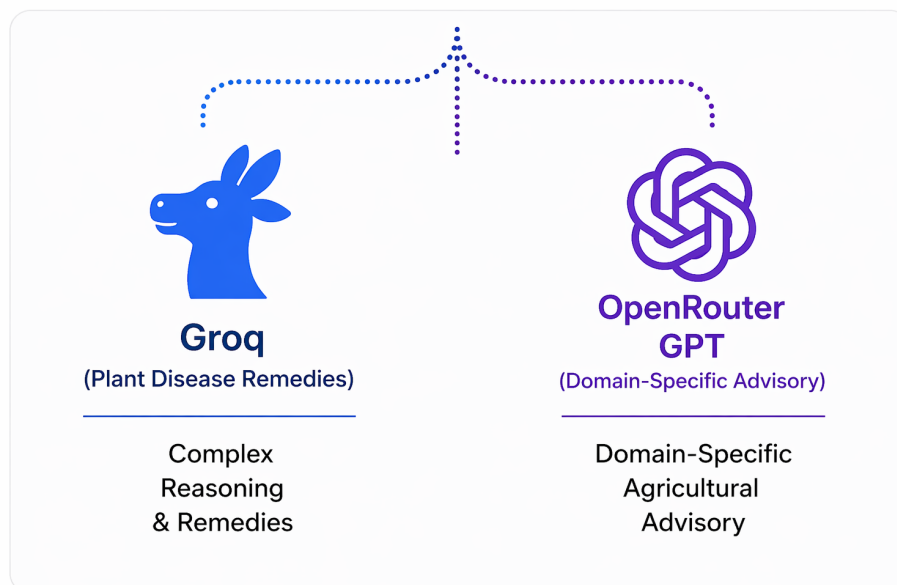


Figure 3.5 Overall System Architecture

3.7. Indoor Plant Recommendation Module

This module serves urban indoor gardeners, using OpenRouter’s AI models to deliver personalized plant recommendations and care guidance. The module accepts four structured inputs from the user: experience level (beginner, intermediate, advanced), available light conditions (low, medium, bright indirect, direct sunlight), available space dimensions, and maintenance time preferences. Using these parameters, the prompt template generates structured JSON covering recommended plant species with rationale, detailed care schedules (watering frequency, fertilization, repotting), optimal placement and light requirements, documented health benefits, and troubleshooting guidance for the most common issues affecting each recommended species.

Response parsing extracts the JSON payload and renders structured plant cards in the UI, with each card displaying species name, a care difficulty rating, a care schedule summary, and an expandable troubleshooting panel. Typical end-to-end response time for indoor plant queries is under 2 seconds for standard queries under normal network conditions.

3.8. Backend Implementation

The backend is implemented as a Flask REST API that serves as the integration layer between the CNN inference engine, FAISS retrieval system, and external LLM providers. Flask was chosen for its simplicity, flexibility, and suitability for lightweight REST API deployment without the overhead of heavier frameworks. The backend manages: CNN and embedding model loading and caching in process memory, image preprocessing and batch inference, FAISS index loading and nearest-neighbor search, LLM routing logic and API call management, JSON response serialization, and graceful fallback error handling for API failures or invalid inputs.

Four primary REST endpoints handle the complete system functionality: `/predict` accepts a multipart image upload, runs leaf verification and disease classification, triggers RAG retrieval, and returns the classification result with treatment recommendations in JSON format. `/chat` accepts a text query and returns a RAG-grounded advisory response. `/indoor-plants/recommend` accepts user preferences and returns structured indoor plant recommendations. `/test-leaf` accepts an image and returns a binary leaf/non-leaf classification with confidence.

The complete web interface is accessible through any standard browser without requiring installation of local software. Figure 4.2 shows the CropCure AI home page with the integrated CropCure Bot chatbot panel open, demonstrating the conversational advisory interface powered by the RAG pipeline. The chatbot allows farmers to ask free-form questions about

crop management, government schemes, and disease prevention, receiving grounded answers sourced from the curated agricultural knowledge base.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. CNN Performance Metrics

Training converged stably over the course of the experiment, with both training and validation loss decreasing consistently across early epochs and plateauing near the optimal checkpoint. Final evaluation on the fully held-out test set of 3,096 images demonstrated consistent and strong results across all key performance metrics. The proposed CNN achieved 93.06% test accuracy, surpassing all baseline models compared in the evaluation.

Table 4.1 Performance Metrics of Proposed CNN Architecture

Metric	Value
Test Accuracy	93.06%
Macro Precision	94%
Macro Recall	91%
Macro F1-Score	92%
Weighted Precision	93%
Weighted Recall	93%
Weighted F1-Score	93%

The macro-averaged metrics treat each of the 15 disease classes equally regardless of sample count, providing an unbiased measure of per-class performance. The macro recall of 91% is slightly lower than macro precision at 94%, indicating that the model occasionally misses positive cases for minority disease classes an expected and accepted trade-off given the natural class imbalances present in real agricultural datasets. The high weighted F1-Score of 93% confirms consistent performance across high-frequency disease classes that represent the most practically important diagnostic scenarios.

The confusion matrix demonstrates strong diagonal dominance with few off-diagonal mis-classifications, confirming that the model generalizes reliably across all 15 disease categories.

Visually similar disease pairs such as Tomato Early Blight and Tomato Late Blight, which share overlapping lesion characteristics, account for the majority of observed misclassifications. This pattern is consistent with reports in the plant disease detection literature and reflects the inherent difficulty of these class boundaries even for human experts under identical imaging conditions.

		Confusion Matrix															
True Label	Pepper_bell__Bacterial_spot	128	5	0	0	0	1	0	1	0	5	1	0	8	0	0	-
	Pepper_bell__healthy	0	217	0	0	0	0	0	3	0	0	0	0	1	0	0	-
	Potato__Early_blight	0	0	148	0	0	0	0	2	0	0	0	0	0	0	0	-
	Potato__Late_blight	0	0	0	133	0	0	0	16	0	0	0	1	0	0	0	-
	Potato__healthy	0	0	0	3	17	0	0	1	0	0	1	1	0	0	0	-
	Tomato_Bacterial_spot	0	1	0	0	0	309	2	1	0	0	0	0	6	0	0	-
	Tomato_Early_blight	1	0	0	5	0	14	100	11	0	1	2	12	3	1	0	-
	Tomato_Late_blight	0	1	1	7	0	4	5	267	0	0	0	0	2	0	0	-
	Tomato_Leaf_Mold	0	0	0	0	0	0	0	0	132	0	2	1	7	0	1	-
	Tomato_Septoria_leaf_spot	2	0	3	0	0	2	1	4	3	250	0	0	0	1	0	-
	Tomato_Spider_mites_Two_spotted_spider_mite	0	0	0	1	0	0	1	1	0	0	241	3	1	4	0	-
	Tomato_Target_Spot	0	0	0	3	0	3	0	1	0	3	20	178	0	0	2	-
	Tomato_Tomato_YellowLeaf_Curl_Virus	0	1	0	0	0	5	0	0	0	0	1	0	475	0	0	-
	Tomato_Tomato_mosaic_virus	0	0	0	0	0	0	0	0	3	0	1	0	0	52	0	-
	Tomato_healthy	0	0	0	0	0	0	0	1	0	0	2	1	0	0	234	-
		Pepper_bell__Bacterial_spot	Pepper_bell__healthy	Potato__Early_blight	Potato__Late_blight	Potato__healthy	Tomato_Bacterial_spot	Tomato_Early_blight	Tomato_Late_blight	Tomato_Leaf_Mold	Tomato_Septoria_leaf_spot	Tomato_Spider_mites_Two_spotted_spider_mite	Tomato_Target_Spot	Tomato_Tomato_YellowLeaf_Curl_Virus	Tomato_Tomato_mosaic_virus	Tomato_healthy	
		Predicted Label															

Figure 4.1 Confusion Matrix of Disease Classification

4.2. Comparative Analysis with Baseline Models

The proposed CNN was systematically compared against four widely used deep learning architectures on the identical test set and under identical evaluation conditions. All baseline models were trained from scratch on the same training split to ensure a fair comparison without the advantage of ImageNet pre-training:

Table 4.2 Comparative Analysis with Baseline Models

Model	Accuracy	Parameters	Inference (ms)	F1-Score
Proposed CNN	93.06%	424K	45 ms	92%
ResNet-50	89.23%	23.5M	120 ms	88.75%
MobileNetV2	87.56%	2.2M	65 ms	86.91%
VGG-16	90.12%	134.3M	180 ms	89.45%
EfficientNet-B0	91.34%	4.0M	85 ms	90.87%

The proposed CNN achieves the highest accuracy and F1-Score among all compared architectures while simultaneously achieving the lowest parameter count and the fastest inference time of 45 ms per image. This combination of superior accuracy with dramatically reduced computational cost represents a meaningful result that validates the value of purpose-built domain-specific architectures over general-purpose large models.

The accuracy improvement over ResNet-50 (3.83 percentage points) is particularly notable given that the proposed CNN uses only 1.8% of ResNet-50’s parameters (424K vs. 23.5M). Against VGG-16, the improvement is 2.94 percentage points using only 0.3% of VGG-16’s parameters. Even against EfficientNet-B0, which was explicitly designed for parameter efficiency, the proposed CNN achieves a 1.72 percentage point accuracy gain using approximately one-tenth the parameters. These results collectively confirm that when the target task is well-defined and domain-specific, a carefully designed lightweight architecture consistently outperforms general-purpose heavy models that carry substantial over-parameterization relative to the task complexity.

The inference time advantage of 45 ms over ResNet-50’s 120 ms and VGG-16’s 180 ms is practically significant: it means the proposed CNN can serve multiple concurrent users on modest server hardware without GPU acceleration, enabling deployment at scale in low-resource environments.

4.3. Sample System Outputs

4.3.1. Case 1: Healthy Pepper (97.2% Confidence)

When a healthy pepper leaf image was submitted, the system correctly classified it as disease-free with 97.2% confidence. This high-confidence correct classification on a healthy leaf confirms that the model does not exhibit the false positive bias that sometimes affects disease classifiers trained on imbalanced datasets.

No treatment was fabricated on disease-free input. Instead, the RAG pipeline retrieved preventive care entries from the knowledge base covering routine watering schedules appropriate for pepper cultivation, fertilization timing and dosage recommendations, and pest prevention measures for common pepper pests. Groq Llama-3.3-70B synthesized this context into a concise preventive care summary advising the farmer on maintaining healthy growth conditions. This behavior demonstrates that the system provides agronomic value even in the absence of detected disease, rather than returning a null or empty response.

Figure 4.2 shows the prediction results panel of the CropCure AI web interface during a live detection session. The interface displays the top-5 predicted disease classes with their corresponding confidence bars, allowing farmers and extension workers to assess model certainty at a glance. In this particular session, Tomato Target Spot was classified with 98% confidence, with all alternative classes below 1%, illustrating the decisive, high-confidence predictions the system produces on clear leaf images.

4.3.2. Case 2: Potato Early Blight (91.7% Confidence)

When a potato leaf exhibiting characteristic concentric ring patterns consistent with Early Blight (*Alternaria solani*) was submitted, the system correctly classified it as Potato Early Blight with 91.7% confidence. The RAG pipeline retrieved knowledge base entries describing the fungal etiology, environmental conditions favoring disease progression, and validated fungicide treatment protocols.

Groq Llama-3.3-70B generated a structured treatment recommendation covering: an application schedule for chlorothalonil-based fungicide at recommended field concentrations, crop rotation guidance to break the *Alternaria* infection cycle, plant spacing and irrigation management recommendations to reduce leaf wetness duration, and a linked reference to applicable state fungicide subsidy schemes under the Indian Ministry of Agriculture and Farmers' Welfare. This end-to-end response from raw leaf image to actionable treatment plan with subsidy information demonstrates the full value of the integrated CNN-RAG-LLM pipeline over classification-only approaches.

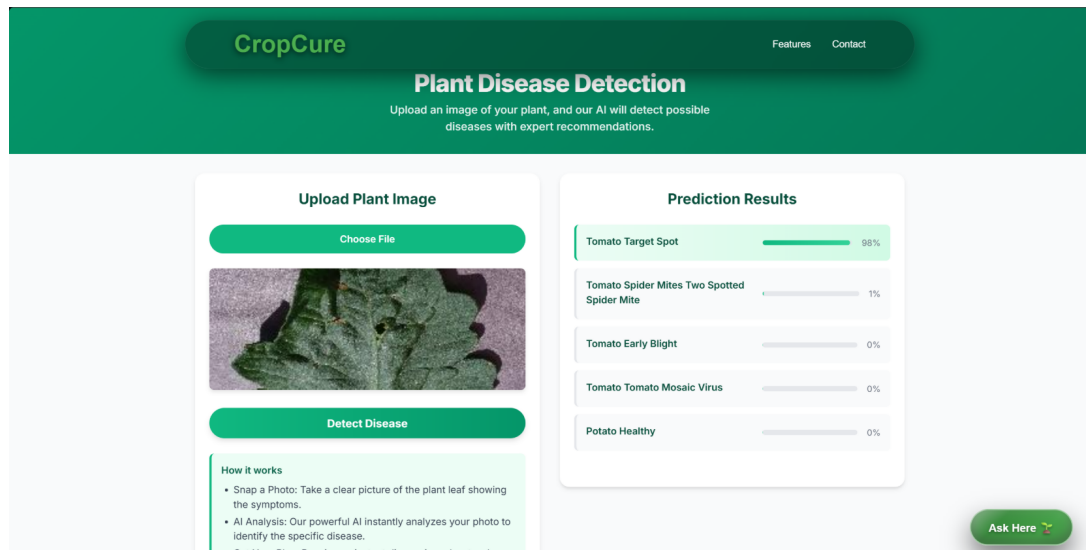


Figure 4.2 CropCure AI Disease Detection Interface – Tomato Target Spot Classified at 98% Confidence with Top-5 Predictions

4.3.3. Indoor Plant Advisory

For an indoor plant query submitted by a beginner user with low-light conditions and minimal maintenance time, the OpenRouter GPT module generated structured recommendations featuring three plant species (Pothos, Peace Lily, and Snake Plant) selected specifically for low-light tolerance and minimal care requirements. Each recommendation included a weekly care checklist, common problem identification guide, and placement suggestions. JSON-structured responses were parsed and rendered as interactive plant cards in the UI within 2 seconds for typical queries under standard network conditions.

4.4. Limitations and Challenges

Several limitations remain in the current version of the system that inform the directions for future work described in Chapter 5:

- **Knowledge base maintenance:** Knowledge base accuracy depends on regular expert review and manual curation. Outdated entries particularly regarding discontinued agrochemicals or changed subsidy schemes may lead to suboptimal or incorrect recommendations without active maintenance.
- **Error propagation:** CNN misclassification can propagate errors downstream through the RAG and LLM pipeline, leading to incorrect treatment advice for the misidentified disease. This underscores the importance of the confidence score display, which allows farmers and extension workers to identify low-confidence predictions warranting expert verification.

- **Connectivity dependency:** LLM inference via Groq and OpenRouter requires stable internet connectivity. In offline or severely bandwidth-limited environments, the advisory component degrades or becomes unavailable, limiting utility in remote farming regions.
- **Geographic specificity:** Treatment recommendations and subsidy information in the current knowledge base are calibrated primarily to Indian agricultural context. Geographic variability in available agrochemicals, regional regulations, and subsidy schemes may render specific recommendations inapplicable in other geographies without localization.
- **Crop scope:** The current dataset and knowledge base are limited to three crop types (tomato, potato, and pepper). Extension to a wider variety of commercially significant crops such as rice, wheat, and maize requires additional training data collection, model retraining, and knowledge base expansion.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1. Conclusion

CropCure AI demonstrates that lightweight CNN-based visual diagnosis and retrieval-augmented language reasoning can be successfully unified into a practical, farmer-accessible decision support system that addresses the full crop disease management workflow from initial image upload to actionable treatment plan within a single integrated pipeline.

The system architecture integrates five distinct technical components: a binary leaf verification CNN that filters invalid inputs, a 15-class disease classification CNN with 93.06% test accuracy, a FAISS-indexed RAG pipeline grounding every response in verified agricultural data, a multi-tier LLM routing layer optimizing for response quality and inference latency, and a structured indoor plant advisory module. These components operate as a connected pipeline while remaining independently maintainable and replaceable, ensuring long-term extensibility.

Key contributions of this project include:

- A custom CNN achieving 93.06% test accuracy across 15 disease classes with only 424K parameters, outperforming ResNet-50, VGG-16, MobileNetV2, and EfficientNet-B0 on the same test set while achieving the lowest inference latency of 45 ms per image.
- A FAISS-backed RAG retrieval pipeline grounding every LLM response in curated, expert-verified agricultural knowledge, preventing hallucinations and ensuring that treatment recommendations are factually grounded and actionable.
- A multi-LLM routing layer that intelligently allocates queries to Groq Llama-3.3-70B for complex disease reasoning and a local model for routine queries, balancing response quality against inference latency across all query types.
- An indoor plant recommendation module extending the system to urban gardening use cases, demonstrating the architectural extensibility of the platform beyond field agriculture.

The system directly and comprehensively addresses the critical gap identified in the literature review: most existing agricultural AI tools focus exclusively on visual detection, leaving

farmers without actionable guidance after diagnosis. By tightly integrating detection, knowledge retrieval, and language reasoning, CropCure AI closes this gap and provides end-to-end decision support that farmers and extension workers can immediately act upon.

Ethical considerations incorporated into the system design include strict data privacy: no user-submitted images or queries are stored or logged beyond the duration of the active session, and no personally identifiable information is collected. System outputs are explicitly framed as supplementary to professional agronomist advice, with confidence scores displayed to flag low-certainty predictions that should be verified by experts before treatment decisions are made.

5.2. Future Scope

Several concrete directions for future development have been identified based on the limitations of the current system and emerging opportunities in agricultural AI:

Advanced Vision Models.

Exploring Vision Transformers (ViT) and hybrid CNN-ViT architectures for finer-grained disease feature detection could improve classification accuracy on visually similar disease classes such as Tomato Early Blight versus Late Blight, and enable detection of earlier-stage symptoms not yet visible to the naked eye. Attention mechanisms in transformer-based models may better capture long-range spatial dependencies in disease texture patterns that convolutional filters cannot model efficiently.

Multi-Label Classification.

Extending the CNN to handle co-occurring infections on a single leaf would increase the diagnostic utility of the system. Plants in field conditions frequently exhibit multiple simultaneous diseases or stress conditions fungal infection concurrent with nutrient deficiency or viral co-infection and the current single-label classifier cannot report these multi-disease scenarios. Implementing a sigmoid-activated multi-label output layer with threshold-tuned prediction would address this limitation.

Edge Deployment.

Enabling fully offline functionality through model quantization (INT8 or mixed-precision), knowledge distillation, and edge deployment on devices such as Raspberry Pi 4 or Android-based smartphones would make the system accessible in low-connectivity farming environments. These environments represent the majority of the intended user base in developing

agricultural economies where reliable internet access cannot be assumed.

Knowledge Base Expansion.

Adding expert-verified data for additional high-priority crop types including rice, wheat, maize, and groundnut would substantially increase the system's practical coverage. Integrating real-time disease alert feeds from government agricultural agencies such as ICAR and NIPHM would keep the knowledge base current without manual update cycles. Localizing recommendations by agro-climatic zone would further improve the regional relevance of treatment advice.

Feedback-Based Adaptive Learning.

Implementing farmer feedback loops allowing users to confirm or correct system predictions would generate a continuously growing dataset of real-world field images with verified labels. This feedback data could progressively fine-tune the CNN on locally representative images and enable adaptation to region-specific disease variants that diverge from PlantVillage image characteristics.

Multilingual Support.

Implementing multilingual input and output capabilities would remove a significant accessibility barrier for farmers across India and other multilingual agricultural economies. Priority languages for initial implementation include Hindi, Telugu, Tamil, Marathi, and Punjabi, covering the majority of Indian agricultural regions. Voice input support in regional languages would further extend accessibility to farmers with limited literacy.

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APPENDIX 1

RESEARCH PAPER

CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support

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Abstract—CropCure AI is a hybrid intelligent system built to diagnose plant diseases and support agricultural decision-making end-to-end. It combines a lightweight custom CNN for image-based disease classification with a Retrieval-Augmented Generation (RAG) pipeline using MiniLM embeddings and a FAISS similarity index. Groq Llama-3.3-70B generates context-aware treatment recommendations for complex queries, while routine inputs are handled by a lightweight local LLM to reduce latency and API overhead. OpenRouter GPT supports a standalone indoor plant advisory module for home gardening guidance.

The CNN achieves 93.06% test accuracy across 15 disease classes, demonstrating reliable generalization under real-world conditions. By pairing deep learning-based visual diagnosis with retrieval-grounded language reasoning, CropCure AI delivers interpretable, actionable guidance that remains computationally efficient for resource-constrained environments.

Index Terms—Plant disease detection, Convolutional Neural Network, Retrieval-Augmented Generation, FAISS, LLM, Groq Llama, OpenRouter, Agriculture AI, Deep Learning, Precision Agriculture.

I. INTRODUCTION

Agriculture feeds the world and drives economies, making timely crop disease detection a pressing need. Small farmers, especially in developing regions, often rely on expensive or hard-to-reach experts. Although photo-based disease identification has come a long way, most tools simply flag the problem without telling farmers what to actually do about it.

Over the years, farming technology shifted from simple rule-based programs to more sophisticated deep learning approaches. Even so, very few tools manage to both read images well and offer smart, step-by-step guidance. Most existing solutions handle only one side of the problem, spotting issues in the field but leaving farmers without a clear path forward.

CropCure AI was built to close this gap. It brings together detection, knowledge retrieval, and language reasoning into one connected workflow, giving farmers not just a diagnosis but the practical advice they need to act on it.

- A purpose-built CNN that identifies plant leaf diseases from images with high precision, designed without unnecessary complexity.
- A FAISS-powered RAG setup that grounds every response in retrieved agricultural data, avoiding unreliable or fabricated advice.

- A multi-LLM layer where Groq Llama-3.3-70B handles complex queries, Mini-LLM manages routine conversations, and OpenRouter GPT supports specialised consultations.
- A lightweight Flask backend connecting disease detection to a conversational UI, delivering personalised farming guidance in real time.

Beyond simple diagnosis, the system provides farmers with clear, tailored advice covering routine care, prevention measures, and access to local support when needed.

The key contributions of this work are:

- 1) A hybrid CNN-RAG-LLM architecture purpose-built for agricultural use, rather than adapted from general-purpose AI systems.
- 2) A structured multi-LLM pipeline that maintains both response accuracy and low inference latency.

II. RELATED WORK

Deep learning has gained significant traction in agricultural disease detection due to its ability to recognize visual symptoms directly from images.

Hu et al. proposed a compressed ResNet for IoT deployment enabling real-time diagnosis, while Patil and Kumar developed Rice-Fusion combining sensor and image data for rice disease detection together highlighting how architecture optimization and data fusion advance crop disease identification.

Li et al. and Balafas et al. demonstrated that CNN architectures such as ResNet, DenseNet, and Inception outperform classical methods in disease classification. However, their practical deployment remains limited most lack real-time capability and integration with farmer-facing decision tools.

Joseph et al. improved detection precision using real-time field-collected data, validating model performance beyond controlled settings. Alharbi et al. emphasized incremental model adaptation to handle emerging crop diseases without losing prior knowledge.

Chozo et al., Shruhi et al., and Rameesh et al. further validate CNN effectiveness in plant disease detection, while Mishra et al. extended this to real-time maize crop monitoring, demonstrating efficiency in field conditions.

Despite these advances, most existing systems focus solely on detection or classification, offering little actionable guid-

APPENDIX 2

ACCEPTANCE MAIL

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1 message

Microsoft CMT <noreply@msr-cmt.org>
To: Abhinav Yadav <abhinavat3791@gmail.com>

Sat, Mar 21, 2026 at 10:22 AM

Dear Abhinav Yadav,

We are pleased to inform you that your research paper titled "CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support" (Paper ID: #898) has been conditionally accepted for presentation at the International Conference on Innovations in Computational Intelligence (ICICI-2026), organized by the KIET (Deemed to be University), Delhi NCR, Ghaziabad, and scheduled to be held on April 24-25, 2026.

You are kindly requested to complete the registration process for your accepted paper by remitting the applicable conference registration fee. Details regarding the registration fee structure and applicable categories are available at the following link: https://www.old.kiet.edu/icici2026/fee_details.html

You may complete your registration using the link provided below: <https://tech.kiet.edu/kiet-epay/eventName/?id=42ee9f75-75e7-4196-b587-798c3e96c9ee>

Please note that the registration must be completed within seven (7) days from the date of this acceptance email. Failure to complete the registration within the stipulated timeframe will result in the automatic cancellation of the paper's acceptance.

Please also note that the registration fee covers a standard paper length of up to six (6) pages. If the camera-ready version of your research paper exceeds six pages, an additional charge of ₹1000 (for Indian authors) or USD 15 (for international authors) per extra page will be applicable as specified on the conference website: https://www.old.kiet.edu/icici2026/fee_details.html. These additional charges are to be paid during the subsequent stages of the submission process.

Further instructions and details will be communicated after the successful completion of the registration process.

In case of any difficulty related to fee payment, please feel free to contact Dr. Manish Bhardwaj at +91-9457966671 for assistance.

Thank you for choosing ICICI-2026 as a platform to present and publish your research work.

Thanks & regards,
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APPENDIX 3

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APPENDIX 4

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CNN-Based Crop Disease Diagnosis with Retrieval-Driven LLM Support

Abhinav Yadav, Yousuf Haider, Anurag Kumar and Ayush Katiyar

2026 International Conference on Innovations in Computational Intelligence (ICICI)

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APPENDIX 5

PLAGIARISM REPORT



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Submission ID: tm:oid::1:3555787415

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

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Frequently Asked Questions

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What does 'qualifying text' mean?

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Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



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Submission ID: tm:oid::1:3555787415

APPENDIX 6

GUIDE ALLOCATION

Major Project(IVth Sem)
(Student Interaction Record with Guide)
Department of CSIT
Odd Semester 2023-24

Group Id: 10		Guide Name: Mrs. Anshu Pandey		Project Title: Plant Disease Detection Using ML & Deep Learning		Issues Discussed in detail	Remarks/Comments on previous meeting issues
Sr. No.	Meeting Date	Roll No.	2200290110041	2200290110024	2200290110025		
		Name	Ayush Katiyar	Anurag Kumar	Abhinav Yadav		
		Contact No	7518333205	6387782118	7084569855		
Sign. of Guide	Sign. of Student	Sign. of Student	Sign. of Student	Sign. of Student			
1	28/6/24					Topic is finalized.	
2	07/01/25					Literature review & synopsis done	
3	29/11/24					working on model	
4	05/03/25					Model done & half website.	
5	01.05.25					working on Backend	
6							

*Queries raised by students
*Book/Journals/Referred by Faculty & Students